

CE 694: Credit Seminar

Credit Seminar Report

Bridging the Mobility Gap: A Study on Demand-Responsive Transit



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Abstract

Public transport in most Indian and global cities is designed around fixed routes and fixed schedules, a model that works well along dense, high-demand corridors but fails in low-density, dispersed, or temporally irregular travel markets. Demand-Responsive Transit (DRT), a mode of public transport in which routes and schedules adapt in real time to passenger requests, has emerged as a way of filling this gap. This report reviews the conceptual foundations of DRT: how it differs from fixed-route transit, how it is classified by operational, functional, technological, economic, and institutional characteristics, and how its global adoption is distributed across regions and institutional models (top-down/public-policy versus bottom-up/private market). It examines the nature of demand in DRT systems, namely temporal, spatial, demographic, and event-driven, and the equilibrium that must be maintained between demand and supply for a DRT system to be viable. The evolution of demand-modelling techniques for DRT is traced through four phases, from static rule-based dial-a-ride formulations of the pre-2010s to the graph-convolutional and deep-learning approaches of the current decade. A detailed case study of the On-Demand Transit (ODT) system in Belleville, Ontario is then presented, in which machine-learning algorithms, namely Random Forest, Bagging, Artificial Neural Networks, and Deep Neural Networks, are used to model daily trip production and distribution, with SHAP (SHapley Additive exPlanations) analysis used to interpret the influence of demographic and trip-based explanatory variables. The report concludes that population density, median income, and the working-age population share are the strongest predictors of ODT demand, and that a genuinely effective DRT system must ultimately couple demand-side forecasting with supply-side (fleet and routing) responsiveness.

1 Introduction

Public transport ridership is shaped by a small number of dominant statistics: cities account for a large and growing share of global population and travel demand, and in India this growth has been accompanied by chronic under-provision of flexible, last-mile-friendly transit options. Conventional fixed-route transit (FRT), buses and trains operating on pre-defined paths and timetables regardless of real-time passenger need, remains efficient in dense, high-frequency corridors but performs poorly in suburban, peri-urban, and off-peak contexts where demand is sparse, dispersed, and irregular.

Demand-Responsive Transit (DRT) is a form of public transport in which vehicle routing and scheduling respond dynamically to passenger requests rather than following a fixed

pattern. It can be summarised by a simple conceptual relationship (Figure 1): DRT combines the flexibility of demand-driven routing with the shared-vehicle, public character of transit.

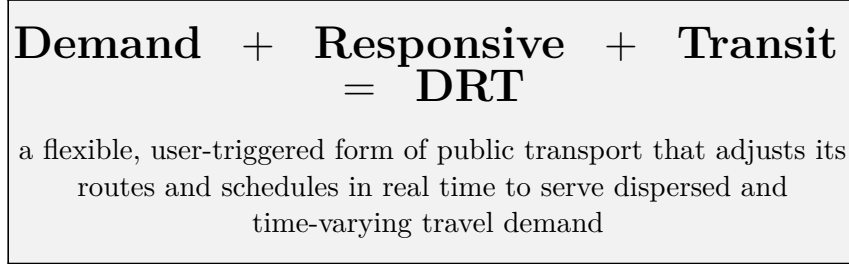


Figure 1: Conceptual definition of Demand-Responsive Transit (DRT)

This report reviews the background, classification, institutional context, and demand-modelling literature associated with DRT, and then examines in detail a published case study of an operational DRT system, the On-Demand Transit (ODT) service in Belleville, Ontario, Canada, in which machine-learning models are used to predict daily trip production and distribution from demographic and trip-based explanatory variables.

2 Background and Objectives of the Study

DRT is not a new idea: dial-a-ride style services managed through phone bookings and radio dispatch date to the 1970s, initially designed to serve low-density or rural areas and users with special mobility needs (Ambrosino et al., 2003). What has changed is the technological substrate: smartphones, GPS, real-time data processing, and digital payments, which has transformed DRT from a niche paratransit service into a scalable option for general public use (Freiberg et al., 2021).

A substantial body of literature has examined DRT from complementary angles: its demand-supply interactions with ride-hailing and conventional public transport (Cats et al., 2022), its long-run sustainability and the reasons many pilot schemes fail to survive beyond the trial stage (Currie & Fournier, 2020), its role in capacity-constrained elastic-demand contexts (Wang et al., 2014), and its potential integration with autonomous vehicles (Djavadian & Chow, 2017; Greenblatt & Shaheen, 2015). Deka et al. (2022) frame DRT development as balancing flexibility against accessibility across a dynamic urban landscape, while Velaga et al. (2012) connect poor transit accessibility directly to transport poverty and its social consequences.

The objectives of this credit seminar are to:

1. Establish a clear conceptual distinction between transit systems and transportation more broadly, and situate DRT within the wider taxonomy of transit modes.

2. Review the operational, functional, technological, economic, and institutional characteristics that define and classify DRT systems, along with their current global distribution.
3. Examine how demand is understood and modelled in DRT systems, and trace the evolution of demand-modelling approaches from static rule-based methods to modern machine-learning techniques.
4. Analyse a detailed, data-driven case study (Belleville, Ontario) of trip production and distribution modelling for an operational ODT system, and draw out its implications for future DRT design.

3 Literature Review

3.1 Transit Systems versus Transportation

A *transit system* is a shared, scheduled mode of public transport (buses, metros, suburban rail) that moves multiple passengers along common corridors, in contrast to *transportation* in the broader sense, which includes private modes such as personal vehicles, taxis, and walking. Table 1 summarises this distinction, and Table 2 illustrates it with reference to Mumbai’s transit ecosystem.

Table 1: Transit System versus Transportation

Aspect	Transit System	Transportation (Broader)
Scope	Shared, scheduled public modes (bus, metro, rail)	All modes of movement, including private and non-motorised
Operation	Fixed or demand-responsive routes serving many users	Individually controlled trips (car, taxi, walk, cycle)
Objective	Mass mobility along common corridors	Point-to-point movement of people/goods

Table 2: Example from Mumbai: Urban vs. Peri-Urban Transit Comparison

Feature	Urban Mumbai (City)	Peri-Urban (Fringe)	Mumbai
Dominant transit modes	Suburban rail, Metro, BEST buses	Shared autos, informal para-transit	
Frequency & coverage	High frequency, dense network	Sparse, irregular coverage	
Typical trip length	Short-to-medium (4–12 km)	Longer, more dispersed	

3.2 Classification of Transit Modes

Transit modes can be classified along three dimensions: their right-of-way (ROW), their system technology, and their type of service. Figure 2 reproduces this classification tree, using Mumbai examples throughout.

Transit Mode (defined by)		
Right-of-Way (ROW)	Category A (Fully Exclusive)	Mumbai Metro Line, Mumbai Local
	Category B (Partially Separated)	Mumbai Monorail
	Category C (Mixed)	Mumbai BEST buses, streetcars
System Technology	Support, Guidance, Propulsion, Control	
Type of Service	By Coverage	Short haul, city transit, regional transit
	By Stop Pattern	Local, accelerated, express
	By Time	Regular (all day), commuter (peak), special

Figure 2: Transit Mode Tree (adapted from source report, Figure 2)

Understanding this structure matters because DRT is not designed to replace traditional transit but to fill the gaps it leaves, particularly in areas or at times with low or scattered demand. DRT can reuse many of the same physical components (depots, power supply, ITS tools) as fixed-route transit, but deploys them more flexibly.

3.3 Transit Travel Characteristics

Effective transit and DRT planning requires understanding not only how many people travel, but *how*, *when*, and *under what conditions*. Three categories of demand are commonly distinguished:

- **Actual demand:** the number of people currently using the service.
- **Potential demand:** people who would use transit if service were better (more frequent, comfortable, or accessible).
- **Latent demand:** the gap between the two, caused by poor accessibility, unaffordable fares, or inadequate service quality.

Key influencing factors include service quality (Level of Service, LOS: speed, frequency, reliability, comfort, cost), urban structure (dense mixed-use areas support transit better than sprawl), inter-modal integration, trip purpose (work trips peak during commute hours; leisure/shopping trips are more dispersed), and trip length (short trips of 4–8 km favour buses/streetcars, medium trips of 6–12 km favour metros, and longer trips favour

suburban/regional rail). Spatially, most cities exhibit a radial pattern toward the Central Business District, though new subcentres (e.g., BKC in Mumbai, Gurgaon near Delhi) are increasingly significant trip attractors. Temporally, transit use shows strong hourly peaks (8–10 a.m. and 5–7 p.m.), weekday/weekend variation, and seasonal effects. Because transit supply (fleet size, staffing) cannot be adjusted as flexibly as demand changes, demand prediction and flexible scheduling, the foundation of DRT, become essential tools for matching supply to real-time demand.

3.4 Key Factors Leading to the Emergence of DRT

The rise of DRT reflects a convergence of structural gaps in traditional transit and enabling technological change. The key contributing factors identified in the literature are:

1. **Spatial demand mismatch:** fixed-route transit concentrates on central corridors, leaving low-density or newly developing areas underserved.
2. **Temporal demand variation:** sharp peaks and troughs that fixed-route systems cannot efficiently scale to match.
3. **Latent and unmet demand:** riders deterred by overcrowding, poor frequency, or lack of nearby access.
4. **Limited first-mile/last-mile connectivity:** particularly from housing societies, narrow lanes, or peri-urban areas to trunk transit stations.
5. **Inflexibility of traditional systems:** fixed routes and schedules regardless of actual passenger needs.
6. **High operational costs in low-demand areas:** large vehicles running near-empty are inefficient; smaller DRT vehicles cost less to operate.
7. **Technological advancements:** mobile apps, GPS, real-time tracking, and digital payments have made DRT operationally viable.
8. **Growth of informal and semi-formal paratransit:** shared autos, taxis, and app-based cabs demonstrate strong latent demand for flexible service.
9. **Passenger behaviour and expectations:** convenience, real-time information, and personalisation, which fixed-route systems cannot offer.
10. **Equity and inclusion gaps:** women, elderly citizens, and low-income groups often have travel needs unmet by conventional systems.
11. **Urban decentralisation and new subcentres:** growth of business hubs outside the traditional downtown (e.g., BKC in Mumbai).
12. **Policy shifts and global trends:** a global push toward sustainable, inclusive, adaptive transport solutions.

3.5 Exploring the Concept of Demand-Responsive Transit

DRT refers to a flexible form of public transportation in which routes and schedules are not fixed but instead adjust in response to real-time passenger demand, typically using small-to-mid-sized vehicles accessed through digital platforms (Freiberg et al., 2021). DRT is often described as the “chameleon” of public transport, capable of taking any form in any environment according to any situation (Volinski, 2019).

3.5.1 DRT Characteristics

DRT systems are defined by characteristics falling into five categories (Figure 3): **operational** (route flexibility, scheduling, vehicle dispatch, hours of operation, booking method), **functional** (first/last-mile connectivity, coverage of underserved areas, coordination with existing transit), **technological** (real-time tracking, booking apps, route optimisation algorithms, data analytics), **economic** (fare structure, cost recovery, funding source, revenue per ride), and **institutional** (regulatory framework, government policy, public-private partnerships, coordination with other modes).

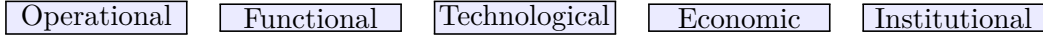


Figure 3: DRT Characteristics: five defining categories

3.5.2 Classification of DRT Service Types

DRT systems are broadly classified into six service types based on target user segment and coverage pattern (Figure 4):

Paratransit Services	Corporate and University Services	Intercity and Inter-region Services
Specific Destination Schemes	Night Services	General Public Services

Figure 4: Types of DRT

- **Paratransit Services:** specialised transport for individuals with disabilities or limited mobility (e.g., Atende+, São Paulo).
- **Corporate and University Services:** on-demand campus/office shuttles (e.g., Shuttll, CityFlo in India).
- **Intercity and Inter-region Services:** flexible-schedule inter-city travel (e.g., Buser, 4bus in Brazil).
- **Specific Destination Schemes:** services connecting to high-demand generators such as airports or hospitals (e.g., SuperShuttle, USA).

- **Night Services:** route-optimised service for reduced nighttime demand (e.g., Night Bus Service, Belleville, Canada).
- **General Public Services:** broadly inclusive urban mobility services without user restriction (e.g., ViaVan, Amsterdam).

DRT occupies a distinctive position on the transport-mode spectrum, more flexible than fixed-route transit but more shared and cost-effective than a private car or taxi (Figure 5).

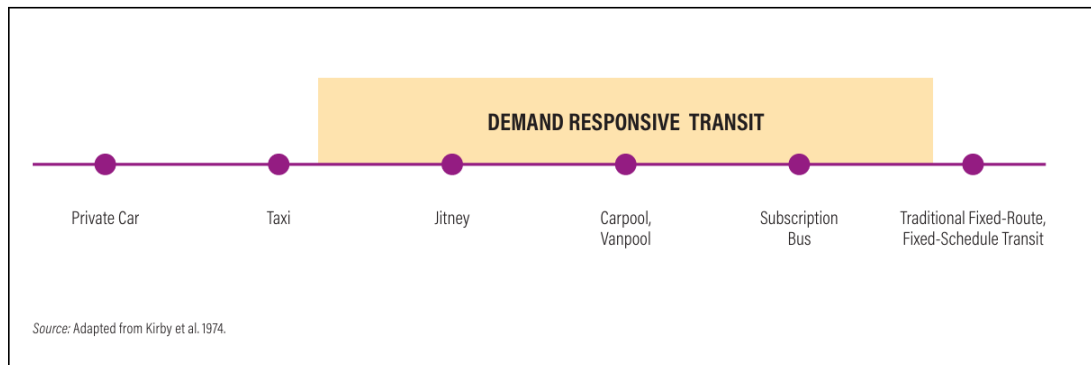


Figure 5: DRT positioned within the transport-mode spectrum (adapted from Kirby et al., 1974)

3.5.3 DRT Attributes

DRT attributes are the flexible, tech-enabled features distinguishing it from fixed-route service: route flexibility, request method, geographical coverage, payment methods, vehicle type, pricing models, real-time operations, booking and reservation method, capacity and efficiency, integration with other transport modes, environmental impact, and level of automation. Route flexibility allows vehicles to adjust paths in real time based on pick-up/drop-off requests, typically booked via mobile apps; pricing is dynamic (distance-, time-, and demand-based); vehicles range from vans and minibuses to electric and autonomous vehicles; and some systems are progressing toward driverless automation to reduce cost and improve operational precision.

3.5.4 Global View of DRT

An initial global survey identified 151 operational DRT cases worldwide. Europe accounts for the largest regional share, while the United States and Australia lead at the country level in systems targeting specific demand segments (Figure 6). Of the 151 systems, 92 serve the general public, 26 serve targeted demand segments (elderly, disabled, students), and 33 are hybrid. DRT adoption remains concentrated in developed countries, with limited presence in India, Brazil, Mexico, Egypt, Bangladesh, and Indonesia, a disparity suggesting significant untapped potential in lower-income regions.

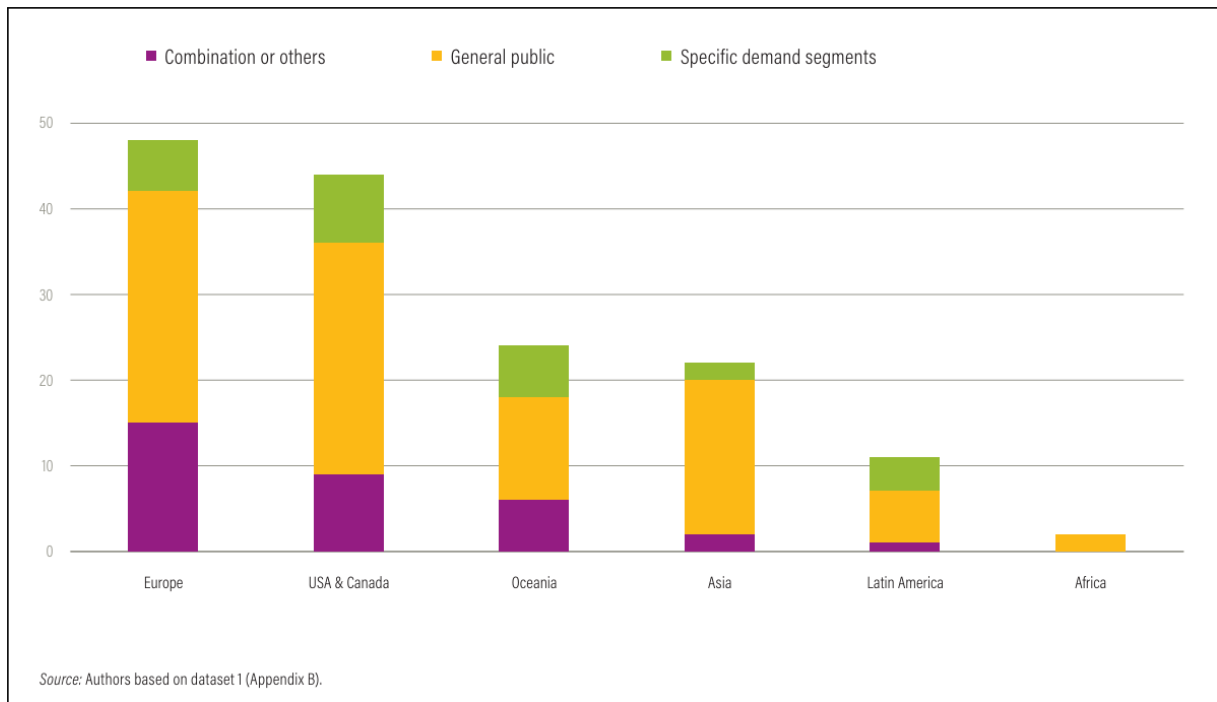


Figure 6: Distribution of sampled DRT types, by region (Source: authors' dataset, Appendix B of source report)

Within India specifically, DRT services identified in the global survey are concentrated in Bengaluru, Chennai, Delhi-NCR, Hyderabad, Kolkata, Mumbai, and Pune, almost all classified as “General Public” services (Table 3). Notably, every identified DRT operator in India is privately owned, with no government-integrated or policy-based DRT system identified, suggesting private companies are filling a market gap that policy has not yet addressed.

Table 3: DRT systems operating in India (Source: authors’ dataset, Appendix B of source report)

#	Region	Country	City	System name	Type of service
1	Africa	Egypt	Cairo	Careem Bus	General Public
2	Africa	Egypt	Cairo	UberBus	General Public
3	Asia	Bangladesh	Dhaka	Pipeep	General Public
4	Asia	India	Bengaluru	ZipGo	General Public
5	Asia	India	Chennai	Shuttl	General Public
6	Asia	India	Delhi-NCR	Shuttl	General Public
7	Asia	India	Hyderabad	Easy Commute	General Public
8	Asia	India	Hyderabad	Shuttl	General Public
9	Asia	India	Kolkata	Shuttl	General Public
10	Asia	India	Mumbai	CityFlo	Combination or others
11	Asia	India	Mumbai	Kruze	General Public
12	Asia	India	Mumbai	Mahindra & Mahindra (Glyd)	General Public
13	Asia	India	Mumbai	Shuttl	General Public
14	Asia	India	Pune	Office Ride	Specific demand segments
15	Asia	India	Pune	Shuttl	General Public

3.5.5 Institutional Framework

Successful DRT implementation depends on the institutional dimension: how the system is organised, regulated, and related to other transport systems. Mulley et al. (2012) distinguish **top-down approaches**, in which government plays the key role in planning and defining services as public policy, from **bottom-up approaches**, in which private operators introduce flexible services in response to market needs. Table 4 compares these two institutional models, and Figure 7 maps three resulting operational models, namely Competitor, Supplementary, and Substitute, against the axes of market/policy origin and integration with public transport.

Table 4: Private Company Services versus Public Policy-Based Systems

	Private Company Services	Public Policy-Based Systems
Purpose	Serve market segments not covered by regular public transit	Planned as part of existing public transport, often as feeder services
Main characteristic	Compete with ride-hailing (Uber) and traditional public transit	Supplement or replace existing transport using advanced technology
Fare system	Usually do not integrate fares with other public transit	Typically integrate fares with existing public transit or offer at no charge
Location	More common in developing countries	More common in developed countries
Examples	Shuttl, CityFlo, Jetty, Uber Bus (India, Mexico, Brazil, Egypt)	TAD IDMF, FlexHop, RéSaTao, Bridj, Keoride (France, Australia, USA)

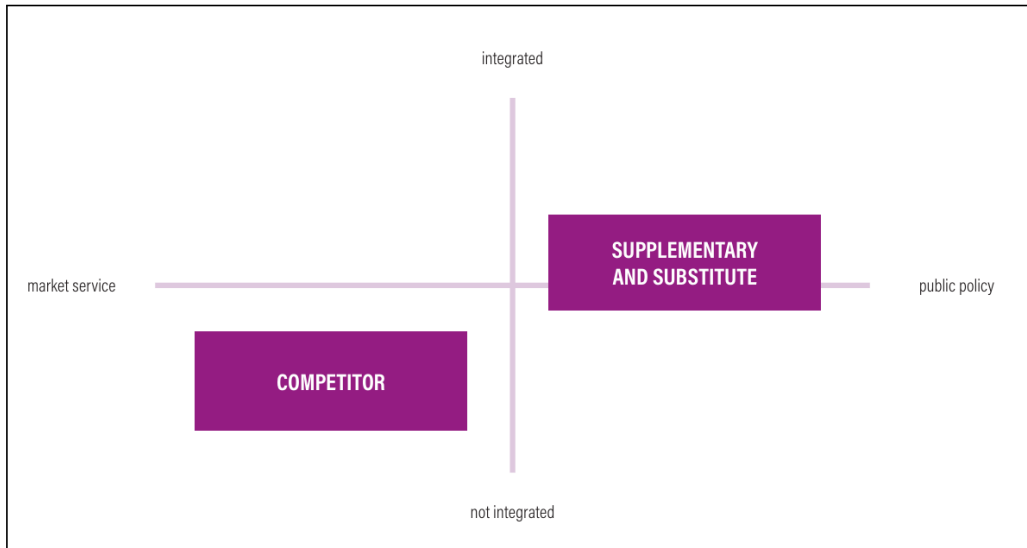


Figure 7: DRT models: institutional origin and integration with public transport

CityFlo in Mumbai exemplifies a blend of the Competitor and Supplementary models: it competes with regular buses and ride-hailing through flexible shuttle services on scheduled routes, while operating with a separate fare system rather than integrating with public transit fares, following the general pattern of private DRT initiatives in other developing countries (Shuttl, Uber Bus).

3.6 Sustainability in DRT and the Dubai Bus-on-Demand Case Study

DRT supports sustainable urban mobility by bridging the gap between the flexibility of private transport and the affordability of public systems. **Environmentally**, it reduces congestion and emissions through shared rides, electric/autonomous vehicles, and real-time route optimisation. **Socially**, it enhances access for underserved populations, namely rural, low-income, elderly, and disabled users, supporting equitable access to jobs, education, and healthcare. **Economically**, sustainability depends on balancing operating costs with fare revenue; the Cost Recovery Ratio (CRR) is a key metric for financial viability.

The **Dubai Bus-on-Demand (DBOD)** service, operating since February 2020 under a Public-Private Partnership, offers a practical illustration of these principles. The service uses 12–14-seat wheelchair-accessible vehicles with fares based on service kilometres. Capacity utilisation increased by 23.5% from 2020 to 2021, with hourly utilisation reaching 8+ riders per vehicle per hour, among the highest of any global micro-transit service. Figure 8 summarises the key operational results.

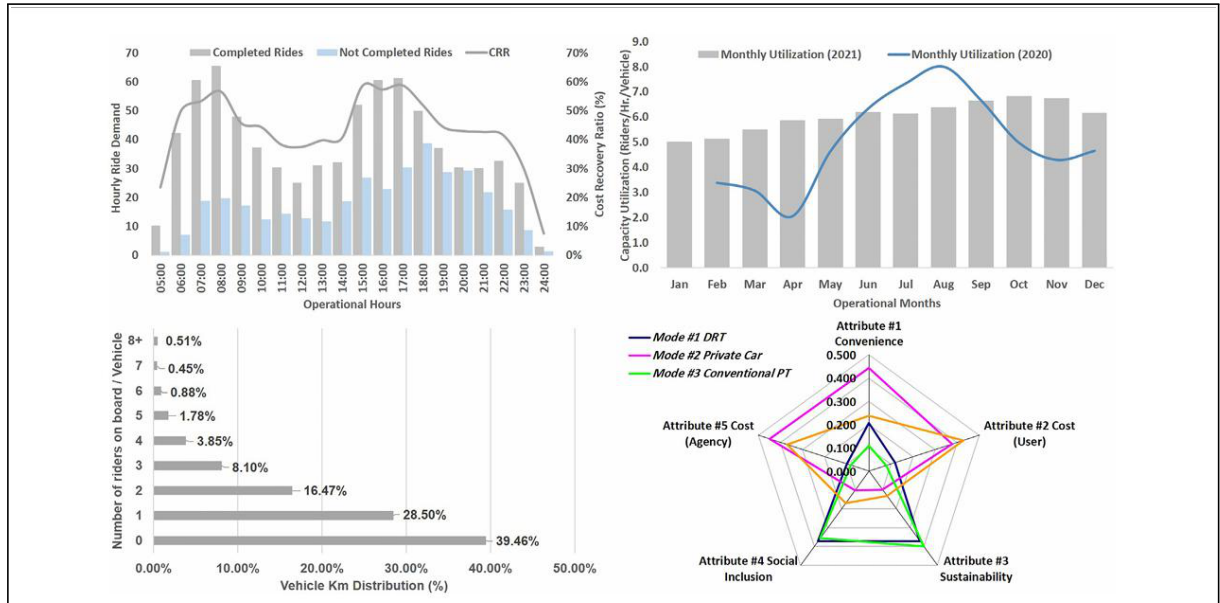


Figure 8: Results from the Dubai Bus-on-Demand case study: hourly ride demand and cost recovery ratio, monthly capacity utilisation (2020 vs. 2021), vehicle-km distribution by occupancy, and comparative mode attributes

Three factors emerged as critical to commercial sustainability: (1) **mode choice**, balancing sustainability and social-inclusion attributes against user-specific needs; (2) **integrated planning for cost recovery**, where CRR is typically higher during high-demand peak hours and struggles during off-peak service; and (3) **service efficiency**, where smaller vehicles (e.g., Toyota Hiaces) improved Expected Time of Arrival and vehicle utilisation

relative to larger buses, which struggled with constraints such as U-turns.

3.7 Challenges and Limitations of DRT

Despite its promise, DRT faces several persistent challenges: a lack of standardised definition or categorisation (making comparative analysis difficult), operational complexity in achieving genuinely effective real-time responsiveness, regulatory and financial integration challenges (particularly in developing regions), and limited comprehensive data for cross-context comparative studies.

3.8 What is Demand in DRT?

Demand in DRT refers to the number, timing, location, and characteristics of trip requests made by users of flexible, on-call transit services. Unlike fixed-route transit, where demand is largely shaped by supply (schedules and routes), DRT demand is dynamic, decentralised, and multi-dimensional, requiring active participation from users who define their own trip origins, destinations, and desired timings (Alsaleh & Farooq, 2021). Table 5 contrasts DRT and fixed-route transit (FRT) across the key planning dimensions.

Table 5: Demand in DRT versus Fixed-Route Transit (FRT)

Aspect	DRT	FRT
Service model	On-demand, dynamically routed	Pre-scheduled, fixed routes/stops
User role	Active: users request rides via apps/calls	Passive: users adjust to fixed routes
Routing	Flexible, real-time adjustment	Static, rarely adjusted
Demand nature	Uncertain, fluctuates in time/space	Predictable, based on historical patterns
System responsiveness	High: responds instantly	Low: delayed by timetable revisions
Service coverage	Can target low-density/underserved areas	Economical only in high-density corridors
Infrastructure need	Light: efficient dispatch, digital booking	Heavy: fixed stops, signage, schedules
Cost efficiency	Efficient when demand is clustered	More cost-efficient in high-demand peaks
Technology use	High: GPS, apps, real-time algorithms	Moderate: mostly scheduling/fare collection
Performance metrics	Wait time, detour length, ride acceptance	Schedule adherence, ridership per trip

3.8.1 Key Aspects of Demand in DRT Systems

1. **Temporal aspects:** demand peaks during morning/evening commutes, with week-day/weekend and seasonal variation requiring dynamic scheduling.
2. **Spatial aspects:** residential zones generate early-morning requests; commercial/institutional areas are high-demand destinations during the day, producing one-way patterns and vehicle repositioning needs.
3. **User demographics and socioeconomic characteristics:** older adults, low-income populations, and car-less households rely more heavily on DRT.
4. **Trip purpose:** regular commuters show predictable patterns; health, leisure, or shopping trips are more spontaneous and dispersed.
5. **Event-driven and external factors:** concerts, sports matches, weather, or transit strikes trigger sudden demand spikes requiring real-time fleet adjustment.
6. **Mode-shift behaviour:** DRT attracts users from other modes when fixed-route transit is inadequate, serving as both a primary mode and a complementary connector.

3.8.2 Assessing Demand Viability

Because a ride request does not automatically signal meaningful or sustainable demand, operators must assess indicators such as repeat usage, duration of demand, trip-purpose diversity, and unmet (latent) demand (Table 6).

Table 6: Indicators for Judging Demand Viability in DRT

Indicator	What It Reveals
Repeat users	Ongoing utility and user satisfaction
Duration of demand	Temporal stability (beyond trial or event-driven usage)
Trip purpose diversity	Coverage of essential needs (work, health, school)
Daily/weekly consistency	Demand regularity across different days and hours
Unmet demand (latent users)	Potential growth if supply or awareness is improved

Low ridership does not necessarily reflect low demand. In the Indian context, suburban connectivity gaps (e.g., Mankhurd–Panvel corridor, where last-mile access to suburban rail is weak and costly), unreliable public bus systems in cities like Patna and Bhopal (where informal e-rickshaws are often cheaper for short trips), and suppressed women’s ridership in rural India (due to safety concerns and poor lighting) all illustrate how ridership figures can significantly understate true transit demand.

3.9 Demand–Supply Equilibrium in DRT

Demand–supply equilibrium is a dynamic state where a DRT system’s transport capacity, namely vehicles, drivers, and routing coverage, aligns with passengers’ travel needs in time, location, and frequency. If supply exceeds demand, idle fleets and unnecessary operating costs result; if demand exceeds supply, wait times lengthen, trips are missed, and satisfaction drops. Table 7 shows how specific demand signals should guide supply responses, and Table 8 summarises the key supply-side components that must be balanced.

Table 7: Aligning Supply with DRT Demand Signals

Observed Demand Pattern	Suggested Supply Response
High elderly population usage	Small vehicles, shorter detours, door-to-door service
Dense, short-trip clusters	Frequent ride pooling, high vehicle turnover
Time-bound work commutes	Scheduled fleet bursts during peak hours
Uneven area demand distribution	Dynamic vehicle repositioning
Night-time safety concerns	Incentivised night shifts or guaranteed availability

Table 8: Key Supply Components in DRT: Role in Balancing Supply

Aspect	Role in Balancing Supply
Fleet size & type	Vehicles (vans, e-rickshaws) matched to demand without overloading during low-use periods
Operating hours	Flexible shifts for early commutes or late-night travel
Dispatching system	GPS-based trip matching and real-time control-centre coordination
Routing algorithms	Smart clustering and re-routing to optimise shared rides, reduce idle time
Fare & incentives	Pricing strategies (discounts, surge pricing) to regulate demand and encourage usage

Achieving this balance relies on mathematical demand-supply models, agent-based simulations of passenger/operator behaviour, and machine-learning techniques that forecast short-term surges from web activity, weather, or local events.

3.10 Evolution of Demand Modelling in DRT Systems: A Four-Phase Analysis

Demand prediction is fundamental to efficient DRT operation: without regional demand predictions, the gap between passengers’ desired and actual boarding times widens

significantly, producing inefficient service with minimal movement and maximum cost. The evolution of demand-modelling approaches can be traced through four phases.

3.10.1 Phase 1: Early Static Prediction Models (Pre-2010s)

The earliest models relied on rule-based algorithms with static assumptions. The Dial-a-Ride Problem (DARP) framework emerged as the fundamental approach for optimising routes under fixed constraints (vehicle capacity, time windows). These models suffered from static demand assumptions, inability to adapt to spatiotemporal variation, limited consideration of passenger-specific needs, and a lack of real-time responsiveness, especially problematic in suburban areas where demand is inherently sparse and heterogeneous.

3.10.2 Phase 2: Statistical Forecasting Era (2010–2018)

This phase saw the application of time-series analysis (ARIMA/ETS models), multi-variable regression, and demand-classification metrics (ADI/CV²). These methods improved temporal forecasting but still struggled with spatial dependencies and heterogeneous demand. As Makridakis et al. observed, statistical methods like ARIMA showed “superiority for stationary urban demand” but “failed for erratic rural demand”, precisely where DRT services are often most needed.

3.10.3 Phase 3: Heuristic and Optimization Models (2018–2022)

The third phase introduced sophisticated multi-objective optimisation approaches. The Search and Rescue Algorithm (SaRA), a heuristic technique inspired by human search-and-rescue behaviour, divides the search space into sub-spaces and assigns “search and rescue workers” to explore them, enabling problem spaces to be effectively divided and conquered and non-promising regions to be efficiently eliminated. Zigrand et al. combined demand clustering with a dynamic DaRP solver, improving insertion rates by 3.5% in suburban areas, demonstrating that geographical/temporal clustering can mitigate data sparsity and that look-ahead optimisation outperforms traditional myopic insertion heuristics.

3.10.4 Phase 4: Machine Learning Revolution (2022–Present)

The most recent phase has seen remarkable advances in machine learning tailored explicitly for DRT demand prediction. The PGDRT model (Prediction Demand based on Graph Convolutional Network) uses both spatial characteristics (proximity between regions, transportation convenience, functional similarity between areas) and temporal characteristics (adjacent visual characteristics, periodic patterns, representative past-demand

features). Because DRT data are inherently sparse due to spatiotemporal effects, Lee et al. employed channel-wise attention and temporal means to alleviate data sparsity before applying ConvLSTM. Other innovations include Deep-DRT (neural networks for multidimensional data), hybrid VAR-CNN models for suburban areas, and spatiotemporal Graph Convolutional Networks that reduced boarding-time gaps by 18%, with some approaches achieving 95% R^2 accuracy despite sparse data.

3.10.5 Critical Challenges and Future Directions

Four challenges recur across the literature: (1) **data sparsity**, where suburban DRT still relies on synthetic data generation to overcome extreme sparsity; (2) **spatiotemporal dependency**, since spatial characteristics affect temporal trends, requiring joint modelling; (3) **equity considerations**, where recent innovations (e.g., DEMO-GRAPH) integrate demographics into demand classification to better serve mobility-restricted passengers; and (4) **autonomous integration**, namely DRT’s potential integration with autonomous vehicles, with projections suggesting up to 40% cost reduction by 2030 via AI-driven fleets.

3.11 Summary of Reviewed Literature

Table 9 summarises the fourteen distinct studies reviewed for this seminar, spanning ride-hailing demand prediction, e-bike/DRT integration, sustainable DRT design, transit-oriented autonomous vehicles, and suburban DRT optimisation frameworks.

Table 9: Summary of Reviewed Literature on DRT Demand Modelling

#	Title / Authors	Yr	Study Area	Key Finding
1	Demand Prediction of Ride-Hailing Pick-Up Location Using Ensemble Learning; Carson-Bell et al.	2021	Chicago, USA	Ensemble/neural models predict demand accurately under privacy constraints; reduced dead-mileage and idle time
2	Combination of e-bike-sharing and DRT in rural areas; Bruzzone et al.	2021	Velenje, Slovenia	Integrated DRT + e-bike-share increased mobility access and was perceived positively by users
3	The journey of demand-responsive transportation: towards sustainable services; Dekka et al.	2023	Dubai, UAE	Cost Recovery Ratio is the most critical sustainability indicator; smaller vehicles increase ETA efficiency

#	Title / Authors	Yr	Study Area	Key Finding
4	DRT for Urban Mobility: Integrating Mobile Data Analytics; Melo, Gomes, Abbasi, Arantes	2024	Porto, Portugal	Data-driven framework reduced stops by 135%, distance by 81%, wait times by 163%, carbon footprint by 73%
5	Real-Time Taxi Demand Prediction Using Data from the Web; Markou, Rodrigues, Pereira	2018	New York City, USA	Fusing event data with time-series models improved prediction (MAE reduced ~37.5%)
6	Transit-oriented autonomous vehicle operation with integrated demand-supply interaction; Wen et al.	2018	Suburban European city	Transit-oriented AV integration enhances availability, supports shared rides, reduces required fleet size by 50%
7	Real-time transit demand prediction capturing station interactions; Noursalehi, Koutsopoulos, Zhao	2018	London Underground (32 stations)	Dynamic factor/ensemble models captured station correlations and event impacts effectively
8	Cloud-Based DRT System Using Forecasting Model; Khair, Dennai, Elmir	2024	(Open taxi data, Kaggle)	Pre-allocation based on predicted demand improved VM utilisation by 25% and cut operating costs
9	Improving demand-responsive transit services: insights from the London field test; Zahedi, Koutsopoulos, Ma	2024	GoSutton DRT pilot, London	Joint DRT-paratransit consolidation cut fleet by 8% and VKT by 13%
10	DRT Services for Improving Public Transport in Suburban Areas; Capodici, Migliore, D'Angelo, D'Orso	2022	Palermo, Italy	DRT-rail integration improves modal share, especially for non-drivers and women; 28.2% more job access
11	Demand Forecasting for Suburban DRT Services; Univ. of Palermo Transport Systems Lab	2022	Palermo, Italy	Spatiotemporal clustering identified efficient zone-formation strategies and fleet performance
12	Optimization-driven Demand Prediction Framework for Suburban DRT Systems; Zigrand et al.	2023	Suburbs of large French cities	Forecast-informed, optimisation-embedded ML gave 3.5% improvement in insertion rate

#	Title / Authors	Yr	Study Area	Key Finding
13	A Multi-task Deep Learning Framework for Sparse DRT Demand; Lee, Choi, Kim	2024	Yeongjongdo, Incheon, Korea	Zero-inflated, multi-task deep learning improved prediction by focusing on sparse OD-pair demand
14	Short-Term Prediction of Demand for Ride-Hailing Services (UberNet); Chen, Thakuriah, Ampoutolas	2021	New York City (Uber data)	CNN-based UberNet outperformed traditional and neural models using embedded spatiotemporal features

4 Case Study: On-Demand Transit in Belleville, Ontario

4.1 Background of the Study

Title of Paper: Interpretable data-driven demand modelling for on-demand transit services

Authors: Nael Alsaleh, Bilal Farooq

Journal: Transportation Research Part A (Policy and Practice), Volume 154, Pages 1–22 (2021)

The Greater Toronto and Hamilton Area (GTHA), home to more than 6.9 million people, is one of Canada’s largest and most populous regions. Approximately 43% of its population resides in dense urban areas, while the remaining 57% live in low-density regions that make up 96% of the total land area, a pattern common to many countries, including India. In these low-density areas, traditional fixed-route public transit (FRT) struggles to operate efficiently: low frequency and limited coverage make service inconvenient, particularly during off-peak hours and in suburban or rural zones.

4.1.1 Research Problem Statement

Traditional fixed-route buses or trains cannot efficiently serve low-demand areas due to high operational costs and vehicle underutilisation during off-peak times, an issue especially acute for shift workers who need transport when conventional transit is scarce. On-demand transit (ODT) has emerged as a cost-effective, flexible, demand-responsive alternative, already implemented in several Canadian cities including Regina, Saskatoon, Calgary, and Chatham-Kent. While these services show promise, there remains a lack of

in-depth research on demand modelling for dedicated-fleet ODT services specifically, the gap this study addresses.

4.1.2 Objectives and Scope

The study develops robust trip-production and trip-distribution models for ODT services, focused on predicting daily trip volumes and understanding how trips are distributed across geographic areas. It investigates how demographic factors (population density, income, working-age population share) and trip characteristics (distance, purpose, time of day) influence demand, using Random Forest, Artificial Neural Networks (ANN), and Deep Neural Networks (DNN), with SHAP analysis used to interpret variable contributions, addressing a gap in interpretable demand modelling tailored specifically to dedicated ODT services.

4.2 Study Area: City of Belleville, Ontario

Belleville is located in Eastern Ontario, about 190 km east of Toronto, and is the largest urban centre in the Quinte region (Figure 9). The city spans 246.8 sq. km with a population density of 205.1 people per sq. km. As per the 2016 Census, Belleville has a population of 50,716, with 63% in the working-age group (15–64 years); median after-tax household income is CAD 53,367, below the Canadian national average of CAD 61,348.

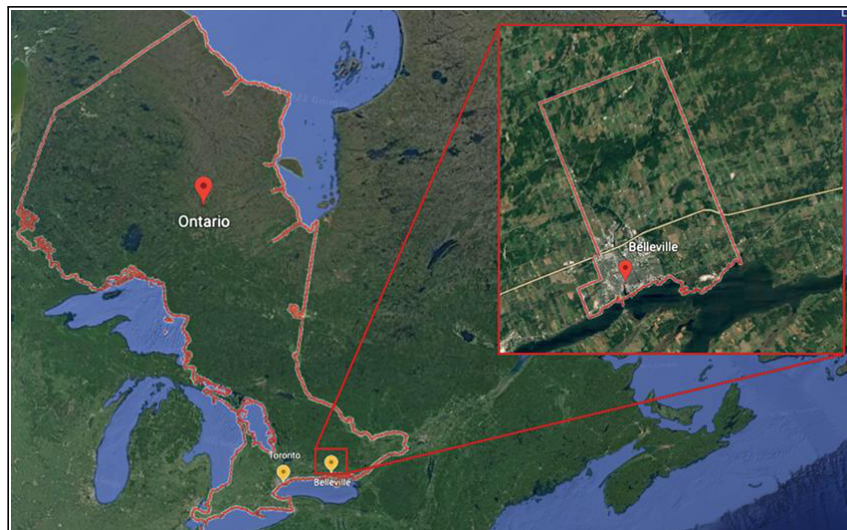


Figure 9: City of Belleville, Ontario, within the Greater Toronto and Hamilton Area

Belleville’s transit system includes 10 fixed routes operated by a fleet of 16 buses. In January 2018, Route 11 was launched as a late-night fixed-route service (9:30 PM–12:30 AM) using two 40-ft buses running every 30 minutes; in September 2018, this service was

converted into an ODT pilot to reduce costs and improve efficiency (Pantonium, 2019; Sanaullah et al., 2021). The ODT service retains the same 40-ft buses, stops, and fare (CAD 3) as the previous fixed-route system, but operates on demand from 9:00 PM to 12:00 AM on weekdays and 7:00 PM to 12:00 AM on weekends, with routing dynamically adjusted based on real-time passenger requests booked through an app, phone, or at-stop boarding. Figure 10 contrasts the fixed and on-demand service footprints.

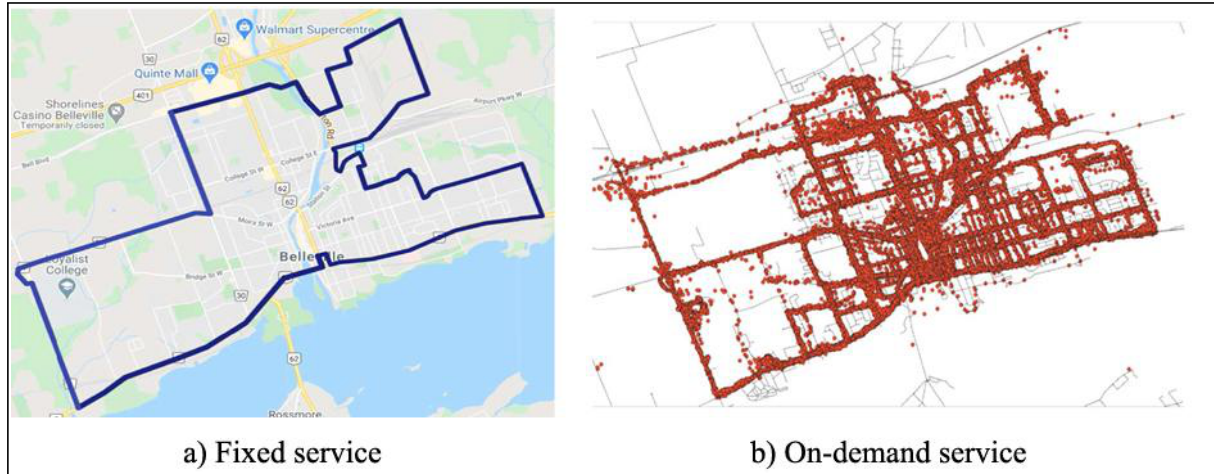


Figure 10: Belleville public transit: (a) fixed-route service coverage; (b) on-demand service pickup/drop-off point density

The transition to ODT resulted in a sustained increase in demand, driven by enhanced flexibility in routing and broader service coverage.

4.3 Methodology and Model Specifications

4.3.1 Data Preparation

The study uses operational data from Belleville’s ODT system collected between 18 September 2018 and 29 May 2019, combined with demographic data from the 2016 Census. The dependent variables are trip production (daily trips generated per dissemination area, DA) and trip distribution (daily trips between area pairs). Explanatory variables span demographic characteristics for both trip origin and destination (population density, median income, average household size, percentage of males, working-age percentage) and trip characteristics (hours of operation per day, day of week, month of year). Table 10 summarises the descriptive statistics for all variables.

Table 10: Descriptive Statistics for Explanatory and Dependent Variables

Variable	Unit	Min	Max	Mean	Std. Dev.
<i>Dependent Variables</i>					
Trips produced per DA per day	count	1	35	4.3	5.29
Trips between any two DAs per day	count	1	11	2.4	1.18
<i>Demographic Characteristics (Origin)</i>					
Population density	people/km ²	63.2	8,139.8	1,700.7	1,628.02
Median income	CAD	24,640	87,296	49,506.2	13,522.32
Average household size	count	1.5	2.8	2.2	0.30
Percentage of males	%	37.2	56.8	47.2	4.24
Working-age percentage	%	38.9	80.7	63.5	8.52
<i>Demographic Characteristics (Destination)</i>					
Population density	people/km ²	63.2	8,139.8	2,075	2,094.04
Median income	CAD	24,640	87,296	46,851.6	13,407.37
Average household size	count	1.5	2.8	2.17	0.25
Percentage of males	%	37.2	56.8	47.5	3.93
Working-age percentage	%	38.9	80.7	66.4	6.54
<i>Trip Characteristics</i>					
Hours of operation per day	binary (0/1)	0	1	–	–
Day of the week	discrete (1–7)	1	7	–	–
Month of the year	discrete (1–9)	1	9	–	–

4.3.2 Model Development

Four machine-learning algorithms, namely Random Forest (RF), Bagging, Artificial Neural Network (ANN), and Deep Neural Network (DNN), were used to model daily trip production and distribution. RF and Bagging are ensemble methods that construct multiple decision trees and combine their outputs; RF selects a random feature subset at each node split, while Bagging uses all available features (Gislason et al., 2006).

4.3.3 Demand Clustering

K-means clustering classified trip production and distribution into three demand clusters based on frequency, namely Low (1 trip/day), Medium (2–5 trips/day), and High (more than 5 trips/day), with the optimal cluster count ($K = 3$) confirmed via the Elbow Method. For trip production, the low-demand cluster was most common (40% of daily trips), followed by medium (36%) and high (24%). For trip distribution, high-level distribution (more than two trips/day between areas) accounted for 39%, medium for 37%, and low for 24%. Figures 11 and 12 show the clustering analysis for production and distribution respectively.

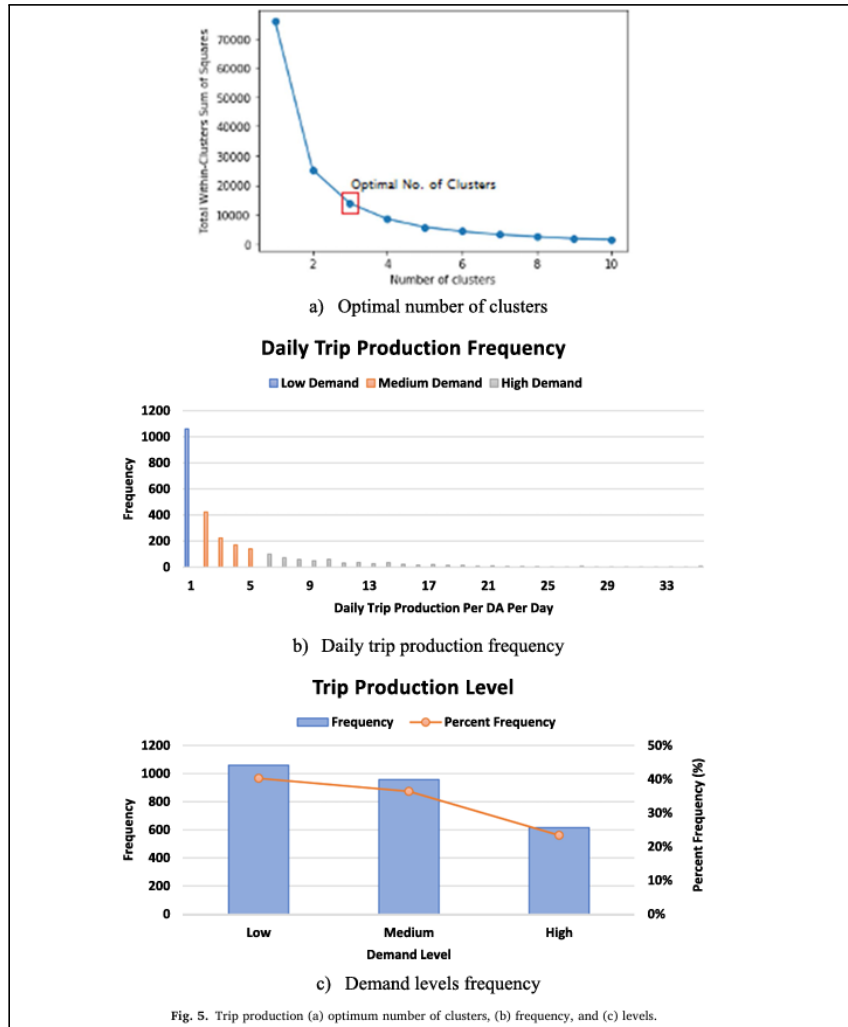


Figure 11: Trip production: (a) optimum number of clusters, (b) daily frequency distribution, (c) demand-level frequency

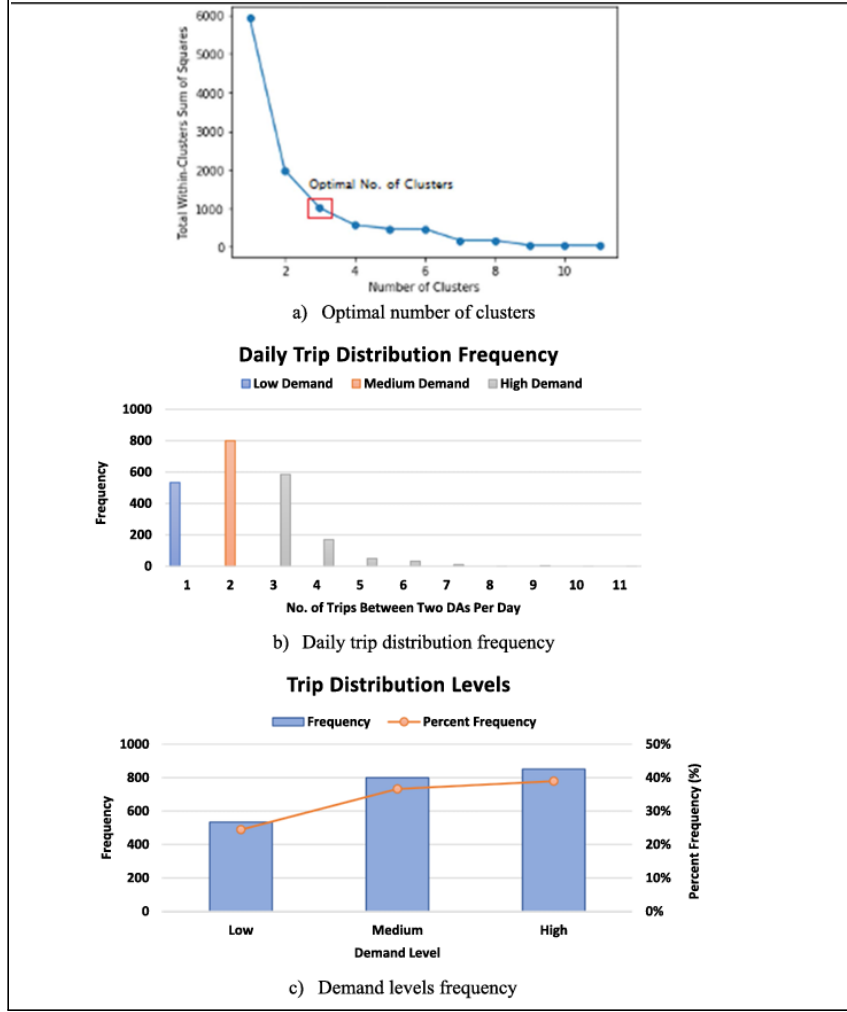


Figure 12: Trip distribution: (a) optimum number of clusters, (b) daily frequency distribution, (c) demand-level frequency

Key takeaways from the clustered data were that demand shows strong temporal variation by time of day and day of week (particularly for high-demand levels), that demographic factors such as population density and income levels significantly influenced demand patterns, and that optimal fleet deployment during peak hours was essential to maintaining service efficiency.

4.3.4 Hyperparameter Tuning and Model Evaluation

Hyperparameters for the four algorithms were tuned using Bayesian Optimization, which reduces the number of evaluations required by intelligently exploring the search space, compared to manual tuning or exhaustive Grid/Random Search (Snoek et al., 2012; Yang & Shami, 2020). Figure 13 lists the hyperparameter sets explored for each algorithm. The dataset was split into training and testing subsets in a 4:1 ratio, with 10-fold cross-validation used during model evaluation to ensure robustness and prevent overfitting.

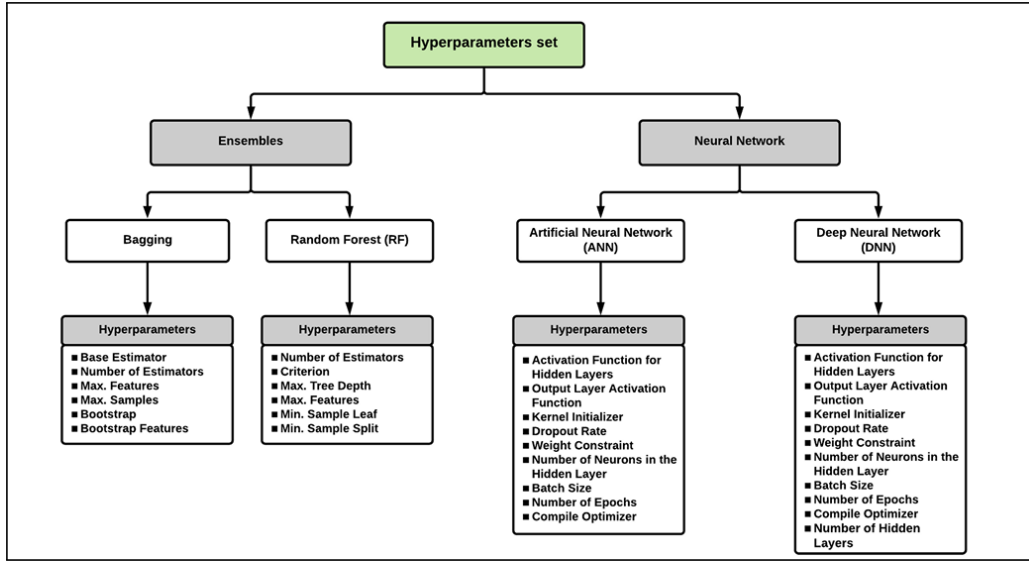


Figure 13: Hyperparameter sets explored for RF, Bagging, ANN, and DNN

Model interpretability was assessed using SHAP (SHapley Additive exPlanations), which explains model outputs by calculating the average marginal contribution of each explanatory variable to the predictions (Lundberg & Lee, 2017).

4.4 Results of the ODT Daily Trip Production and Distribution Models

Figure 14 compares the predictive accuracy of RF, Bagging, ANN, and DNN across the low, medium, and high demand levels for the trip production model; all four algorithms show a consistent dip in accuracy for the medium-demand level, reflecting boundary misclassification between adjacent demand classes.

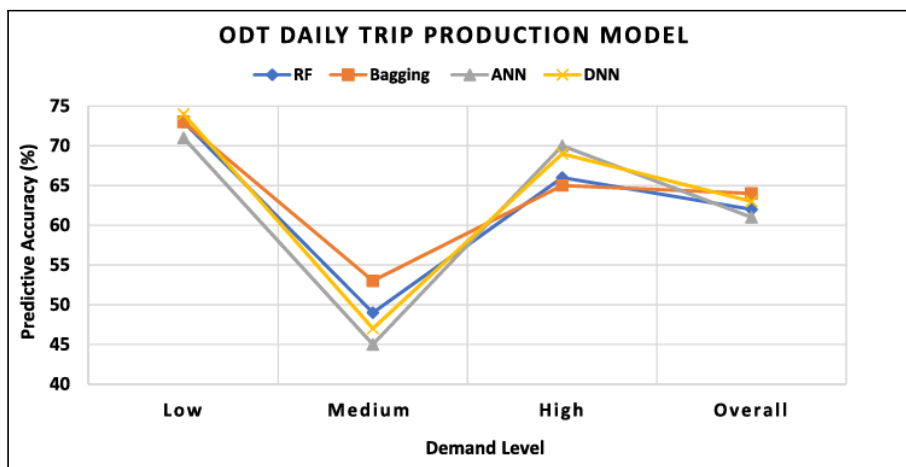


Figure 14: Comparing the predictive accuracy of the ODT daily trip production models

Table 11 summarises algorithm-wise performance for the trip production model, and Table 12 gives the detailed confusion matrix for the best-performing algorithm (Bagging).

Table 11: Trip Production Model: Algorithm Performance Comparison

Algorithm	Acc. Low	Acc. Medium	Acc. High	Overall	Challenges
DNN	74%	45%	60%	64%	Medium-level misclassification (2 trips/day boundary)
Bagging	68%	53%	61%	64%	Lower accuracy for medium production level
ANN	72%	50%	70%	62%	–
Random Forest	70%	51%	65%	63%	–

Table 12: Trip Production Model Testing Confusion Matrix (Bagging Algorithm)

Predicted →	Low	Medium	High	Total	Acc. (%)
Low Demand	107	3	3	113	64
Medium Demand	32	109	27	168	65
High Demand	14	44	98	156	70
Total	153	156	128	437	64

For trip distribution, Table 13 shows RF was the best overall performer (72% accuracy), while DNN achieved perfect accuracy (100%) at the low-demand level.

Table 13: Trip Distribution Model: Algorithm Performance Comparison

Algorithm	Acc. Low	Acc. Medium	Acc. High	Overall
DNN	100%	61%	60%	71%
Random Forest	70%	65%	63%	72%
Bagging	64%	59%	61%	69%
ANN	68%	60%	62%	69%

Table 14: Trip Distribution Model Testing Confusion Matrix (Random Forest Algorithm)

Predicted →	Low	Medium	High	Total	Acc. (%)
Low Demand	107	3	3	113	95
Medium Demand	32	109	27	168	65
High Demand	14	44	98	156	63
Total	153	156	128	437	72

SHAP analysis (Figure 15) identified population density as the single most influential explanatory variable for both models, followed by percentage of males, median income, and working-age percentage; day-of-week and month-of-year were the least influential (Table 15).

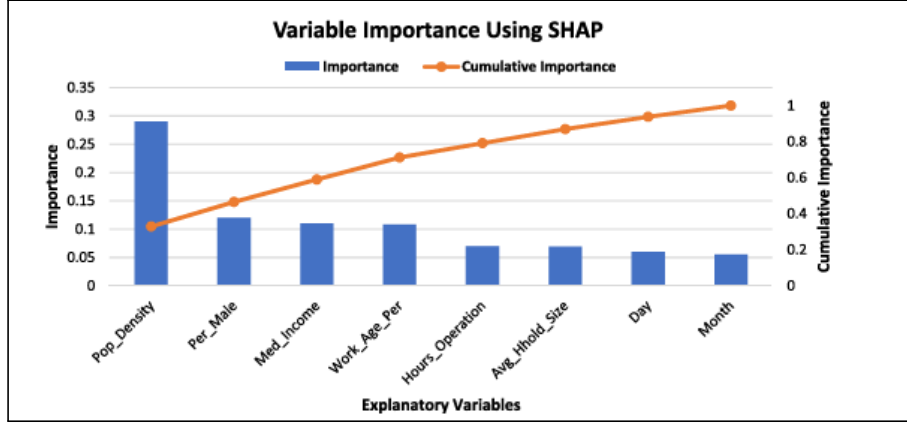


Figure 15: Variable importance using SHAP for the trip production model

Table 15: SHAP Analysis: Variable Importance Ranking

Explanatory Variable	Importance
Population density	Highest
Median income	High
Percentage of males	High
Average household size	Moderate
Working-age percentage	Moderate
Day-of-week	Least
Month-of-year	Least

Key findings: the Bagging algorithm performed best for ODT daily trip production (64% overall accuracy); RF performed best for ODT daily trip distribution (72% overall accuracy); population density, median income, and working-age percentage were the most influential variables for both production and distribution; and DNN was most accurate at predicting the low-demand class in both models (74% for production, 100% for distribution).

4.5 Discussion and Inference

The study reinforces prior findings that land-use and demographic characteristics significantly affect demand for on-demand mobility services (Jain et al., 2017; Ghaffar et al., 2020), with ODT demand higher in commercial/industrial land-use areas, consistent with Sanaullah et al. (2021). It also reveals a behavioural distinction between dedicated

ODT fleets and crowdsourced mobility services: while ride-sourcing services are commonly used for non-work-related activities, ODT users in Belleville primarily use the service for home-to-work or home-to-shop trips. Whereas traditional statistical methods (e.g., gravity models) are commonly used for trip production and distribution modelling, this study demonstrates that machine-learning algorithms such as Random Forest and Bagging offer greater flexibility and accuracy (Pouerebrahim et al., 2019); classification algorithms were used in preference to regression due to the characteristically low, discrete demand values observed for ODT services.

5 Conclusion

This seminar has traced Demand-Responsive Transit from its conceptual origins as a flexible complement to fixed-route transit, through its operational, functional, technological, economic, and institutional classification, to its current global distribution, concentrated in developed countries and, within India, almost exclusively delivered through unregulated private operators rather than public policy. The evolution of demand-modelling techniques across four phases (static rule-based methods, statistical forecasting, heuristic optimisation, and the current machine-learning revolution) demonstrates how far DRT demand prediction has advanced, while unresolved challenges of data sparsity, spatiotemporal dependency, equity, and autonomous integration remain fertile ground for future research.

The Belleville, Ontario case study demonstrates in concrete terms how machine-learning models, namely Random Forest, Bagging, ANN, and DNN, can be used to predict daily ODT trip production and distribution, and how SHAP analysis can make such models interpretable for planners: population density, median income, and working-age population share emerge as the dominant predictors of demand. Cities like those in India, where public transport often misses out on large groups of people, stand to benefit substantially from such interpretable, data-driven demand models. Future work should aim to connect the demand side (where and when people need rides) with the supply side (how many vehicles are needed and where), creating a system that automatically adjusts to real-time travel needs and makes transport more efficient, inclusive, and reliable for all users.

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